

AI/ML MASTER ROADMAP CHECKLIST

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1. Python Fundamentals

Topic	Learn	Project	Revision
Variables			
Data Types			
Type Casting			
Operators			
User Input			
Strings			
String Methods			
Lists			
Tuples			
Sets			
Dictionaries			
If Else			
Match Case			
For Loop			
While Loop			
Functions			
Lambda			
Recursion			
OOP			
File Handling			
Exception Handling			
Modules			
Virtual Environment			
Pip			

2. Mathematics

Topic	Learn	Project	Revision
Scalars			
Vectors			
Matrices			
Matrix Operations			
Dot Product			
Cross Product			
Eigenvalues			
Eigenvectors			
SVD			

Calculus			
Topic	Learn	Project	Revision
Limits			
Derivatives			
Chain Rule			
Partial Derivatives			
Gradients			
Gradient Descent			
Jacobian			
Hessian			

Probability			
Topic	Learn	Project	Revision
Sample Space			
Probability Rules			
Conditional Probability			
Bayes Theorem			
Random Variables			
PMF (Probability Mass Function)			
PDF (Probability Density Function)			
CDF (Cumulative Distribution Function)			
Expectation			
Variance			
Covariance			

Statistics			
Topic	Learn	Project	Revision
Mean			
Median			
Mode			
Standard Deviation			
Variance			
Correlation			
Skewness			
Kurtosis			
Z Score			
Confidence Interval			
Hypothesis Testing			
T-Test			
Chi Square Test			
ANOVA			

3. Data Analysis

Topic	Learn	Project	Revision
NumPy			
Broadcasting			
Array Manipulation			
Vectorization			
Random Module			
Pandas Series			
DataFrames			
Merge			
Join			
GroupBy			
Pivot			
Missing Values			
Feature Engineering			
Matplotlib			
Seaborn			
Plotly			

4. Data Preprocessing

Topic	Learn	Project	Revision
Data Collection			
Data Cleaning			
Duplicate Removal			
Missing Values			
Outlier Detection			
Label Encoding			
One Hot Encoding			
Ordinal Encoding			
Feature Scaling			
Standardization			
Normalization			
Train Test Split			
Pipeline			
Data Leakage			
Feature Selection			
PCA (Principal Component Analysis)			

5. Machine Learning

Topic	Learn	Project	Revision
Supervised Learning			
Unsupervised Learning			
Semi-Supervised Learning			
Self-Supervised Learning			
Reinforcement Learning			

Supervised Learning			
Regression			
Topic	Learn	Project	Revision
Linear Regression			
Multiple Regression			
Polynomial Regression			
Ridge Regression			
Lasso Regression			
ElasticNet			
Support Vector Regression (SVR)			
Decision Tree Regression			
Random Forest Regression			
Gradient Boosting Regression			
XGBoost Regression			
LightGBM Regression			
CatBoost Regression			

Classification			
Topic	Learn	Project	Revision
Logistic Regression			
K-Nearest Neighbors (KNN)			
Naive Bayes			
Support Vector Machine (SVM)			
Decision Tree			
Random Forest			
AdaBoost			
Gradient Boosting			
XGBoost			
LightGBM			
CatBoost			

Unsupervised Learning			
1. Clustering			
Topic	Learn	Project	Revision
K-Means Clustering			
Hierarchical Clustering			
DBSCAN			
Mean Shift			
Gaussian Mixture Model (GMM)			

2. Dimensionality Reduction			
Topic	Learn	Project	Revision
PCA (Principal Component Analysis)			
LDA (Linear Discriminant Analysis)			
t-SNE			
UMAP			
AutoEncoder			

3. Association Rule Learning			
Topic	Learn	Project	Revision
Apriori Algorithm			
FP-Growth Algorithm			

Model Evaluation & Hyperparameter Tuning			
Topic	Learn	Project	Revision
Accuracy			
Precision			
Recall			
F1 Score			
ROC Curve			
AUC (Area Under Curve)			
Confusion Matrix			
Cross Validation			
Grid Search			
Random Search			
Bayesian Optimization			

6. Deep Learning

Topic	Learn	Project	Revision
Perceptron			
Artificial Neural Network (ANN)			
Activation Functions			
Backpropagation			
Loss Functions			
Optimizers			
Batch Normalization			
Dropout			
Regularization			

Convolutional Neural Networks (CNN)			
Topic	Learn	Project	Revision
Convolution			
Pooling			
Padding			
Stride			
ResNet			
VGG			
EfficientNet			
MobileNet			

Recurrent Neural Networks (RNN)			
Topic	Learn	Project	Revision
Vanilla RNN			
LSTM (Long Short-Term Memory)			
GRU (Gated Recurrent Unit)			
Seq2Seq (Sequence-to-Sequence)			
Attention Mechanism			

Transformers			
Topic	Learn	Project	Revision
Self-Attention			
Multi-Head Attention			
Positional Encoding			
Encoder			
Decoder			
BERT			
GPT			
T5			
Vision Transformer (ViT)			

7. Computer Vision

Topic	Learn	Project	Revision
OpenCV			
Image Processing			
Object Detection			
YOLO			
SSD (Single Shot Detector)			
Faster R-CNN			
Image Segmentation			
OCR (Optical Character Recognition)			
Face Recognition			
Pose Estimation			
Object Tracking			
Image Captioning			

8. Natural Language Processing (NLP)

Topic	Learn	Project	Revision
Text Cleaning			
Tokenization			
Lemmatization			
Stemming			
Stopwords Removal			
TF-IDF			
Word2Vec			
FastText			
GloVe			
BERT			
GPT			
Text Summarization			
Machine Translation			
Sentiment Analysis			
Named Entity Recognition (NER)			

9. Generative AI

Topic	Learn	Project	Revision
Prompt Engineering			
Embeddings			
Vector Database			
RAG (Retrieval-Augmented Generation)			
AI Agents			
Function Calling			
MCP (Model Context Protocol)			
LangChain			
LangGraph			
LlamaIndex			
CrewAI			
AutoGen			
DSPy			
Guardrails			
Memory			
Evaluation			

10. Large Language Models (LLMs)

Topic	Learn	Project	Revision
Tokenization			
Context Window			
Fine-Tuning			
LoRA (Low-Rank Adaptation)			
QLoRA			
PEFT (Parameter-Efficient Fine-Tuning)			
RLHF (Reinforcement Learning from Human Feedback)			
DPO (Direct Preference Optimization)			
Quantization			
Distillation			
Hallucination			
Context Engineering			
Speculative Decoding			
KV Cache			

11. MLOps

Topic	Learn	Project	Revision
Git			
GitHub			
Docker			
Kubernetes			
MLflow			
DVC (Data Version Control)			
Apache Airflow			
FastAPI			
Flask			
Streamlit			
CI/CD			
Model Monitoring			
Model Serving			

12. AI Deployment

Topic	Learn	Project	Revision
Flask Deployment			
FastAPI Deployment			
Streamlit Deployment			
Gradio Deployment			
Docker Deployment			
AWS (Amazon Web Services)			
GCP (Google Cloud Platform)			
Microsoft Azure			
Hugging Face Spaces			
Vercel			
Render			

13. Projects

Topic	Learn	Project	Revision
Beginner Projects			
Intermediate Projects			
Advanced Machine Learning Projects			
Deep Learning Projects			
Natural Language Processing (NLP) Projects			
Computer Vision Projects			
Large Language Model (LLM) Projects			
RAG (Retrieval-Augmented Generation) Projects			
Agentic AI Projects			
MLOps Projects			

14. Research

Topic	Learn	Project	Revision
Reading Research Papers			
arXiv			
Reproducing Research Papers			
Benchmarking			
Experiment Tracking			
Writing Research Reports			

15. Interview Preparation

Topic	Learn	Project	Revision
Machine Learning Interview Questions			
Deep Learning Interview Questions			
Statistics			
Python			
SQL			
Case Studies			
System Design			
Coding Practice			